TEAM VAANAM

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SLA-based Bandwidth Verification for ILL Services

TANFINET HACKATHON 2026

PROBLEM STATEMENT

Government departments rely on Internet Leased Line (ILL) services to support applications such as e-governance platforms, record systems, and public service delivery. These services operate under Service Level Agreements (SLAs) that define guaranteed bandwidth, uptime, latency, packet loss, and restoration timelines.

There is currently no standardized, independent, and auditable mechanism available to continuously verify whether the delivered bandwidth and network performance meet the agreed SLA parameters.

SOLUTION

Secure, Real-Time SLA Bandwidth Monitoring & Compliance Platform

We propose a secure, scalable SLA Monitoring and Compliance Platform that deploys lightweight Python-based virtual agents (containerized via Docker) across government offices to perform active testing using iPerf3 and ICMP, passive monitoring via SNMP, and cryptographic signing using PKI (RSA), with encrypted transmission to a centralized backend built on FastAPI, a time-series database (Prometheus), and a PostgreSQL audit store; the system processes data through a real-time analytics engine, applies automated SLA scoring algorithms, and visualizes compliance metrics on a Vue 3 dashboard. This delivers tamper-evident logs, automated alerts, digitally signed PDF reports, and regulator-ready audit trails for transparent, data-driven ISP accountability.

Most departments depend solely on ISP-generated reports, which:

- Lack independent verification
- Do not provide real-time visibility
- Are not always audit-ready
- Often cannot create automated reports

There is a need for a secure, reliable, scalable, and tamper-proof bandwidth measurement and reporting system that can:

1. Continuously monitor actual bandwidth performance
2. Measure latency, packet loss, and uptime
3. Compare real performance against SLA commitments
4. Detect and flag SLA violations automatically

KEY FEATURES

Redefining Features

Real-Time Bandwidth Monitoring – Continuously measures throughput, latency, jitter, and packet loss

SLA Compliance Engine – Compares live perf against SLA thresholds and flags violations

Secure Data Transmission – Encrypts and digitally signs all measurement data for tamper-proof reporting

Automated Alerting System – Triggers instant notifs for degradation or SLA breaches

Audit-Ready Reporting – Generates digitally signed monthly compliance reports and violation summaries

Centralized Performance Dashboard – Provides live visualization, ISP comparison, and historical trend analytics

TECHNOLOGY

The Technology Stack

Vue 3

Delivers a stable and scalable frontend for creating custom dashboards that are accessible and user-friendly.

Docker Containers

Ensures consistent, portable deployment of monitoring agents and backend services across environments with simplified scaling and maintenance.

FastAPI

Powers a high-performance backend API layer for secure data ingestion, SLA evaluation, and real-time analytics processing.

Prometheus

Enables efficient time-series data collection and monitoring, supporting real-time metric aggregation and alert generation.

THANK YOU

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